

PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Mathematical Model of Microwave Transmission

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Concept of Wave Modes	2
1.1	Classification of Modes	2
1.2	Field Lines in Waveguide Cross-Sections	2
2	Proof of Non-Existence of TEM Mode in Hollow Waveguides	3
2.0.1	Mathematical Proof	3
3	Losses Associated with Microwave Transmission	4
3.1	Conductor Losses (Skin Effect)	4
3.2	Dielectric Losses	4
3.3	Radiation Losses	5
4	Concept of Wave Impedance	5

1. Concept of Wave Modes

When electromagnetic waves propagate along a guide structure, the boundaries dictate how the fields are oriented. These field distributions are classified as modes.

Wave Mode: A specific configuration of electric and magnetic fields that satisfies Maxwell's equations and the boundary conditions of the transmission structure. Modes are characterized by their cutoff frequencies and spatial patterns.

1.1 Classification of Modes

We classify electromagnetic modes based on the presence of longitudinal field components along the direction of propagation (assumed to be the z -axis):

- **TEM (Transverse Electromagnetic) Mode:** Both the electric and magnetic fields are entirely transverse to the direction of propagation.

$$E_z = 0 \quad \text{and} \quad H_z = 0 \quad (1)$$

- **TE (Transverse Electric) Mode:** The electric field is completely transverse to the direction of propagation, while the magnetic field has a longitudinal component.

$$E_z = 0 \quad \text{and} \quad H_z \neq 0 \quad (2)$$

- **TM (Transverse Magnetic) Mode:** The magnetic field is completely transverse to the direction of propagation, while the electric field has a longitudinal component.

$$H_z = 0 \quad \text{and} \quad E_z \neq 0 \quad (3)$$

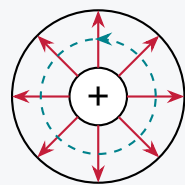
- **Hybrid Modes (HE/EH):** Both electric and magnetic fields have non-zero longitudinal components.

$$E_z \neq 0 \quad \text{and} \quad H_z \neq 0 \quad (4)$$

1.2 Field Lines in Waveguide Cross-Sections

The physical layout of the electric ($\vec{E}$) and magnetic ($\vec{H}$) fields is visualized below:

Field Distribution in Coaxial (TEM) and Rectangular Waveguide (TE_{10})



Coaxial (TEM)

— E-Field

- - - H-Field



Rectangular Waveguide (TE_{10})

2. Proof of Non-Existence of TEM Mode in Hollow Waveguides

A central mathematical model shows that TEM waves cannot propagate inside single-conductor hollow waveguides.

Theorem

TEM waves cannot exist inside a hollow waveguide formed by a single closed conductor. They require at least two distinct conductors (e.g., coaxial line) to propagate.

2.0.1 Mathematical Proof

Assume a TEM wave propagates along the z -direction in a hollow waveguide with a single conductor wall C . For a TEM mode, by definition:

$$E_z = 0 \quad \text{and} \quad H_z = 0$$

From Maxwell's curl equation for the electric field:

$$\nabla \times \vec{E} = -j\omega\mu\vec{H}$$

Analyzing the z -component of the curl:

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z \quad (5)$$

Since $H_z = 0$, this yields:

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0 \quad (6)$$

This indicates that the transverse electric field $\vec{E}_t = E_x \hat{a}_x + E_y \hat{a}_y$ has a curl of zero in the transverse plane. Therefore, the transverse electric field can be expressed as the gradient of a scalar electrostatic potential function $\Phi(x, y)$:

$$\vec{E}_t = -\nabla_t \Phi(x, y) \quad (7)$$

Next, we invoke Gauss's Law in the charge-free dielectric region inside the waveguide:

$$\nabla \cdot \vec{E} = 0 \implies \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

Since $E_z = 0$, the divergence equation reduces to the transverse plane:

$$\nabla_t \cdot \vec{E}_t = 0 \quad (8)$$

Substituting $\vec{E}_t = -\nabla_t \Phi$ into the divergence equation yields:

$$\nabla_t^2 \Phi(x, y) = 0 \quad (\text{Laplace's Equation}) \quad (9)$$

Now we apply the boundary conditions on the waveguide boundary C . Since the wall is a Perfect Electric Conductor (PEC), the tangential electric field must vanish on C . This implies the conductor boundary is an equipotential surface:

$$\Phi(x, y) = \Phi_0 \quad (\text{constant}) \quad \text{on boundary } C \quad (10)$$

According to the **uniqueness theorem** of Laplace's equation: If a potential Φ satisfies Laplace's equation ($\nabla^2\Phi = 0$) within a closed region bounded by a single conductor at a constant potential Φ_0 , then the potential must be constant everywhere inside that region:

$$\Phi(x, y) = \Phi_0 \quad \text{for all } x, y \text{ inside } C \quad (11)$$

Taking the gradient of a constant potential to find the electric field:

$$\vec{E}_t = -\nabla_t\Phi_0 = 0 \quad (12)$$

If the electric field is zero everywhere inside the waveguide, then by Maxwell's equations, the magnetic field is also zero:

$$\vec{H}_t = 0 \quad (13)$$

Thus, no electromagnetic fields can exist inside the guide. Therefore, **TEM modes cannot exist inside a hollow, single-conductor waveguide** (Q.E.D.).

3. Losses Associated with Microwave Transmission

Energy propagating down a microwave line attenuates due to three primary physical loss mechanisms:

3.1 Conductor Losses (Skin Effect)

At microwave frequencies, current does not distribute uniformly across the conductor cross-section but crowds near the surface.

Skin Depth (δ): The depth into a conductor at which the electromagnetic wave amplitude decays to $1/e$ (approx. 36.8%) of its surface value.

Skin Depth and Surface Resistance

The skin depth δ is given by:

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} \quad (\text{m}) \quad (14)$$

The corresponding surface resistance R_s is:

$$R_s = \frac{1}{\sigma \delta} = \sqrt{\frac{\pi f \mu}{\sigma}} \quad (\Omega/\text{square}) \quad (15)$$

Where:

f = frequency of operation (Hz)

μ = permeability of the conductor (H/m)

σ = conductivity of the metal (S/m)

3.2 Dielectric Losses

Dielectric losses occur in the medium filling the waveguide due to the rotation of polar molecules under high-frequency alternating electric fields. It is quantified using the loss

tangent ($\tan \delta$):

$$\alpha_d = \frac{\omega \epsilon''}{2\beta} \approx \frac{\pi \sqrt{\epsilon_r} \tan \delta}{\lambda_0} \quad (\text{Np/m}) \quad (16)$$

3.3 Radiation Losses

Radiation losses occur primarily in open-boundary lines (e.g., Microstrip lines) where fields are not completely enclosed, allowing some energy to radiate outward as radio waves.

4. Concept of Wave Impedance

Wave Impedance (Z_w): The ratio of the transverse electric field to the transverse magnetic field for a propagating mode. It represents the characteristic resistance that the wave encounters as it travels.

The wave impedance depends heavily on the mode type (TE or TM) and the operating frequency:

Wave Impedance Formulations

Let $\eta = \sqrt{\mu/\epsilon}$ be the intrinsic impedance of the dielectric medium. The wave impedances for TE and TM modes are:

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \quad (\Omega) \quad (17)$$

$$Z_{\text{TM}} = \eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} \quad (\Omega) \quad (18)$$

Where:

f_c = Cutoff frequency of the specific mode

f = Operating frequency

Critical Limits:

- For $f < f_c$ (below cutoff): Impedance is purely imaginary, indicating evanescent (non-propagating) fields.
- For $f \rightarrow \infty$: Both Z_{TE} and Z_{TM} approach the intrinsic impedance η of the medium.

Quick Revision — Module II

Mathematical Model of Microwave Transmission | PE-EC701A

Key Formulas at a Glance

Skin Depth	$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$
Surface Resistance	$R_s = \sqrt{\frac{\pi f \mu}{\sigma}}$
TE Wave Impedance	$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}$
TM Wave Impedance	$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$
Intrinsic Impedance	$\eta = \sqrt{\frac{\mu}{\epsilon}} \approx 377 \Omega$ (in free space)

One-Line Definitions

- **Wave Mode:** A valid mathematical field pattern satisfying boundary conditions.
- **TEM Mode:** A mode with zero electric and magnetic fields in the direction of propagation.
- **Wave Impedance:** The ratio of transverse electric field to transverse magnetic field.
- **Skin Depth:** The distance in a metal where current density drops to 36.8% of surface value.
- **Uniqueness Theorem:** Electrostatic theorem proving Laplace's solution is unique under boundary limits.
- **Evanescent Wave:** An attenuated wave that does not carry real power, occurring below cutoff.
- **Loss Tangent:** A measure of energy dissipation in a lossy dielectric material.

Frequently Asked / Viva Questions

Q1. Why can TEM waves not exist inside a hollow waveguide?

Because the waveguide wall is a single conductor that acts as an equipotential boundary. Inside, Laplace's equation forces the potential to be constant, making the electric fields zero.

Q2. What transmission structures support TEM waves?

Structures with two or more isolated conductors, such as coaxial lines, striplines, and microstrip lines.

Q3. How does skin depth vary with frequency?

It is inversely proportional to the square root of frequency ($\delta \propto 1/\sqrt{f}$). Higher frequencies yield thinner skin depths.

Q4. Why do conductor losses increase at microwave frequencies?

Frequently Asked / Viva Questions (contd.)

Because current is confined to a very thin layer (skin depth), effectively reducing the cross-sectional area of conduction and increasing resistance.

Q5. Explain the behaviour of Z_{TE} as frequency approaches the cutoff frequency f_c from above.

As $f \rightarrow f_c^+$, the denominator $\sqrt{1 - (f_c/f)^2} \rightarrow 0$, causing the wave impedance Z_{TE} to approach infinity.

Q6. What happens to the wave impedance of a TM mode below cutoff ($f < f_c$)?

The term inside the radical becomes negative, making the wave impedance purely reactive/imaginary, indicating no real power propagation.

Q7. What is the physical significance of the loss tangent ($\tan \delta$)?

It represents the ratio of the lossy conduction current density to the lossless displacement current density in a dielectric.

Q8. Why do we use copper or silver plating inside microwave guides?

To reduce conductor losses. Since currents flow within a few microns of the surface, a thin layer of highly conductive material reduces surface resistance R_s .

Q9. State the intrinsic impedance of free space.

$$\eta_0 = \sqrt{\mu_0/\epsilon_0} \approx 120\pi \approx 377 \Omega.$$

Q10. What is the difference between TE and TM modes?

In TE modes, the longitudinal electric field is zero ($E_z = 0, H_z \neq 0$). In TM modes, the longitudinal magnetic field is zero ($H_z = 0, E_z \neq 0$).

Q11. How does the cutoff frequency affect mode propagation?

Waves only propagate if their operating frequency exceeds the cutoff frequency ($f > f_c$). Below cutoff, waves decay exponentially.

Q12. What are hybrid modes?

Modes where both the longitudinal electric and magnetic fields are non-zero ($E_z \neq 0$ and $H_z \neq 0$). They are common in optical fibers.

Comparison of Wave Propagation Modes

Feature	TEM Mode	TE Mode	TM Mode
Longitudinal E_z	$E_z = 0$	$E_z = 0$	$E_z \neq 0$
Longitudinal H_z	$H_z = 0$	$H_z \neq 0$	$H_z = 0$
Hollow Guide?	No	Yes	Yes
Wave Impedance	$Z_w = \eta$	$Z_w = \eta / \sqrt{1 - (f_c/f)^2}$	$Z_w = \eta \sqrt{1 - (f_c/f)^2}$
Cutoff Freq f_c	$f_c = 0$	$f_c > 0$	$f_c > 0$